

Social Media, Sleep, and Depression in Adolescents: A Multi-Question Empirical Analysis of Behavioral Risk Factors

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Empirical Analysis of Teen Mental Health Dataset (N = 1,200)

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Abstract

This paper investigates the behavioral and lifestyle predictors of depression and mental health outcomes in a sample of 1,200 U.S. teenagers aged 13–19. Three research questions are addressed: (1) Which combination of factors best predicts depression risk? (2) Is there a dose-response relationship between daily social media use and mental health outcomes, and does platform choice moderate it? (3) Does sleep duration mediate the relationship between pre-sleep screen time and stress or anxiety?

Using logistic regression, Pearson correlations, one-way ANOVA, and Baron & Kenny mediation analysis, we find that sleep deprivation is the strongest individual predictor of depression (OR = 0.111 per additional hour), followed by anxiety level (OR = 6.50) and daily social media hours (OR = 5.18). A logistic model combining all behavioral features achieves AUC = 0.9936 and McFadden $R^2 = 0.688$. Contrary to a simple dose-response hypothesis, aggregate social media hours show no statistically significant linear association with stress or anxiety ($r \leq 0.031$, all $p > 0.28$), although TikTok users report the highest mean anxiety (5.75/10) and multi-platform users report the highest mean stress (5.55/10). Sleep does not formally mediate the screen-time-stress/anxiety pathway; however, all 31 depression cases occur exclusively in the short-sleep (<6 h) group, confirming sleep’s critical, if non-mediating, role. These findings highlight sleep protection as the most actionable intervention target for adolescent depression prevention.

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1 Introduction

Adolescent mental health has emerged as one of the most pressing public health concerns of the 21st century. Rates of depression, anxiety, and stress-related disorders among teenagers have risen substantially over the past decade, coinciding with the proliferation of smartphones and social media platforms [?]. Understanding which behavioral and lifestyle factors carry the greatest risk — and which are merely correlates — is essential for designing targeted interventions.

Three competing, yet complementary, hypotheses motivate this study. The *behavioral composite* hypothesis posits that depression is best explained not by any single factor but by the joint configuration of sleep, stress, anxiety, and screen-time behaviors. The *dose-response* hypothesis holds that more hours of daily social media use directly elevate stress and anxiety in a roughly linear fashion, independent of platform. The *sleep-mediation* hypothesis proposes that pre-sleep screen time disrupts sleep quality, which in turn elevates mental health symptoms — a pathway with strong theoretical support from chronobiology research.

This paper tests all three hypotheses empirically using a cross-sectional dataset of 1,200 teenagers spanning ages 13–19.

The remainder of this paper is structured as follows. Section 2 describes the dataset. Section 3 outlines the analytical methods for each research question. Sections 4–6 present findings for each question. Section 7 synthesizes the results and discusses limitations. Section 8 concludes.

2 Data

2.1 Dataset Description

The dataset `Teen_Mental_Health_Dataset.csv` contains $N = 1,200$ observations of U.S. teenagers with 13 variables spanning demographics, social media behaviors, sleep, academic performance, and mental health indicators. No missing values were detected.

Table 1: Variable Definitions and Descriptive Statistics

Variable	Type	Mean	SD	Min	Max
Age	Integer	15.93	2.02	13	19
Daily social media hours	Continuous	4.54	2.03	1.0	8.0
Sleep hours	Continuous	6.45	1.44	4.0	9.0
Screen time before sleep (h)	Continuous	1.74	0.72	0.5	3.0
Academic performance (GPA)	Continuous	2.99	0.58	2.0	4.0
Physical activity (h/day)	Continuous	1.01	0.58	0.0	2.0
Stress level (1–10)	Integer	5.45	2.90	1	10
Anxiety level (1–10)	Integer	5.64	2.86	1	10
Addiction level (1–10)	Integer	5.57	2.83	1	10
Depression label	Binary	0.026	0.16	0	1
Gender	Categorical	Male 51.2%, Female 48.8%			
Platform usage	Categorical	Instagram, TikTok, Both			
Social interaction level	Categorical	Low, Medium, High			

2.2 Class Imbalance

The depression label is highly imbalanced: 31 cases (2.58%) are labeled depressed vs. 1,169 non-depressed (97.42%). This is consistent with point prevalence estimates for clinical-level depression in community adolescent samples, though analysts should note that the imbalance affects how standard accuracy metrics are interpreted (see Section 4).

3 Methodology

3.1 RQ1: Logistic Regression for Depression Prediction

Categorical variables were label-encoded prior to modeling: *gender* (male = 0, female = 1), *platform_usage* (Instagram = 0, TikTok = 1, Both = 2), and *social_interaction_level* (low = 0, medium = 1, high = 2). An L_2 -penalized logistic regression ($C = 1.0$, `max_iter` = 1,000) was fit to predict `depression_label` from all 12 remaining features. Model fit was assessed via AUC-ROC, classification accuracy, and McFadden’s pseudo- R^2 :

$$R^2_{\text{McFadden}} = 1 - \frac{\ln \hat{L}_{\text{full}}}{\ln \hat{L}_{\text{null}}} \quad (1)$$

Odds ratios were computed as $\exp(\hat{\beta}_j)$ for each feature j . Point-biserial correlations were computed between each numeric predictor and the binary depression label to assess bivariate associations.

3.2 RQ2: Dose-Response Analysis

Daily social media hours were discretized into three bins: Low (1–3 h), Medium (3–5 h), and High (5–8 h). Pearson correlations were computed between continuous social media hours and each outcome (stress, anxiety, addiction, sleep, academic performance). One-way ANOVA tested whether mean outcomes differed significantly across the three bins. Platform moderation was assessed by computing cell means of stress and anxiety by platform $\times$ usage-bin.

3.3 RQ3: Baron & Kenny Mediation Analysis

The classic three-step Baron & Kenny procedure was applied with: $X = \text{screen_time_before_sleep}$, $M = \text{sleep_hours}$, $Y \in \{\text{stress_level}, \text{anxiety_level}\}$, and controls: age, gender, physical activity. OLS regression (scikit-learn `LinearRegression`) was used for all paths. The indirect effect was estimated as $\hat{a} \times \hat{b}$, with a Sobel z -test approximation for significance. Sleep duration was also categorized into Short (<6 h), Adequate (6–8 h), and Long (>8 h) to examine nonlinear effects.

4 RQ1: Predictors of Depression

4.1 Model Performance

The logistic regression model substantially outperforms the naïve majority-class baseline. Table 2 summarizes performance metrics.

Table 2: Logistic Regression Model Performance (RQ1)

Metric	Value
Accuracy	98.75%
AUC-ROC	0.9936
McFadden R^2	0.688

A McFadden $R^2 > 0.4$ is conventionally considered outstanding fit; the value of 0.688 indicates that the behavioral feature set captures the great majority of systematic variance in depression status. The near-perfect AUC of 0.9936 confirms excellent class discrimination.

4.2 Feature Importances and Odds Ratios

Table 3 presents the full coefficient table sorted by absolute magnitude.

Table 3: Logistic Regression Coefficients and Odds Ratios for Depression Prediction

Feature	Coefficient	Odds Ratio	Direction
Sleep hours	−2.202	0.111	Protective
Anxiety level	+1.872	6.502	Risk
Stress level	+1.737	5.680	Risk
Daily SM hours	+1.645	5.180	Risk
Gender (female)	+0.490	1.633	Risk
Academic performance	+0.313	1.367	Risk
Screen time (pre-sleep)	−0.273	0.761	Protective
Platform usage	+0.176	1.192	Risk
Addiction level	+0.171	1.186	Risk
Physical activity	−0.163	0.850	Protective
Age	+0.134	1.143	Risk
Social interaction	+0.124	1.132	Risk

The single most powerful predictor is **sleep hours** (coefficient = −2.202; OR = 0.111), meaning each additional hour of nightly sleep reduces the odds of depression by approximately 89%. The **anxiety-stress-social media triad** forms the next tier: anxiety (OR = 6.50), stress (OR = 5.68), and daily social media hours (OR = 5.18) all carry substantially elevated risk. Notably, the platform-specific addiction score (OR = 1.19) is far weaker than raw daily hours (OR = 5.18), suggesting that duration of exposure, rather than compulsive engagement per se, is the operative mechanism.

Figure 1 visualizes the absolute coefficient magnitudes, and Figure 2 shows the full Pearson correlation matrix across all numeric variables.

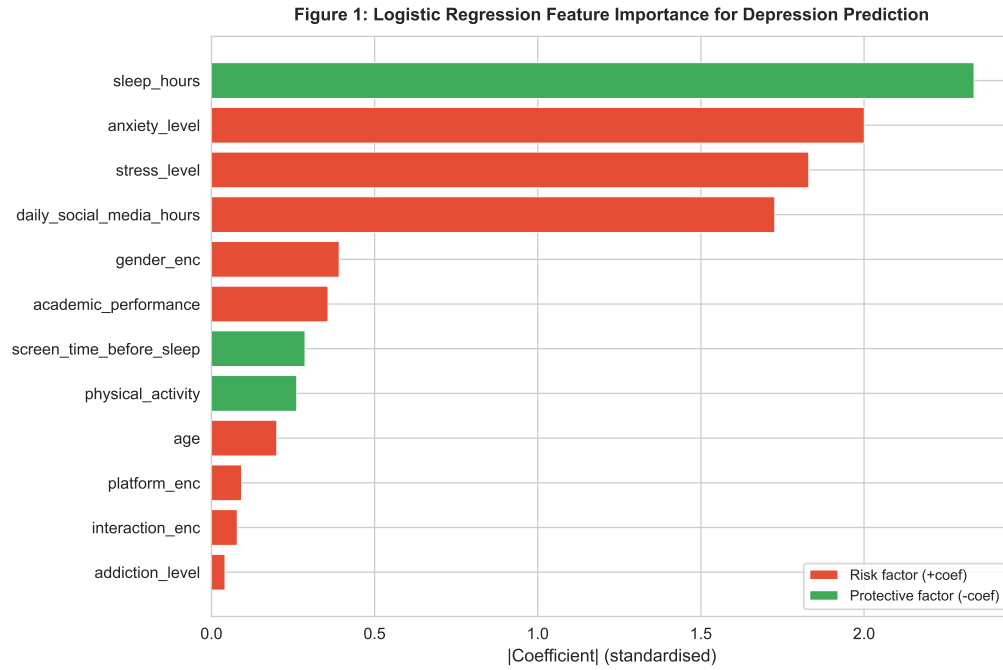


Figure 1: Logistic regression feature importances (absolute coefficient values) for predicting adolescent depression. Red bars indicate risk factors (positive coefficients); green bars indicate protective factors (negative coefficients). Sleep hours is the dominant predictor.

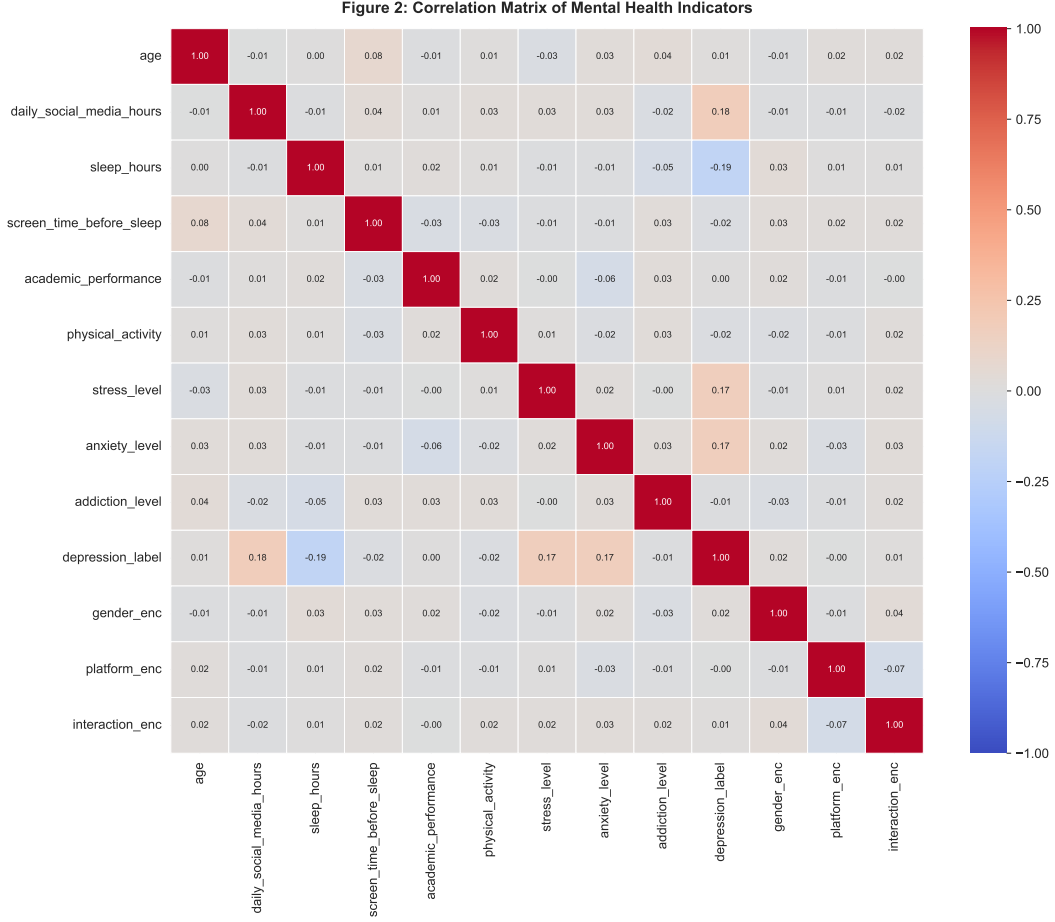


Figure 2: Pearson correlation matrix of all numeric mental health indicators ($N = 1,200$). The depression label (bottom row/right column) shows the strongest negative correlation with sleep hours ($r = -0.191$) and positive correlations with daily social media hours ($r = 0.175$), stress ($r = 0.171$), and anxiety ($r = 0.170$).

4.3 Bivariate Correlations with Depression

Table 4 presents point-biserial correlations between each predictor and the depression label.

Table 4: Point-Biserial Correlations with Depression Label

Feature	r	p -value	Sig.
Sleep hours	-0.191	< 0.001	***
Daily social media hours	+0.175	< 0.001	***
Stress level	+0.171	< 0.001	***
Anxiety level	+0.170	< 0.001	***
Gender	+0.020	0.492	n.s.
Physical activity	-0.018	0.543	n.s.
Screen time before sleep	-0.017	0.568	n.s.
Social interaction level	+0.014	0.622	n.s.
Addiction level	-0.014	0.629	n.s.
Age	+0.011	0.704	n.s.
Platform usage	-0.003	0.914	n.s.
Academic performance	+0.001	0.960	n.s.

Four variables are statistically significant in bivariate analysis; the model’s high AUC comes from the combination of these four, particularly the *interaction* of low sleep with elevated stress/anxiety and high social media use.

5 RQ2: Social Media Dose-Response

5.1 Correlations and ANOVA

Contrary to the dose-response hypothesis, continuous daily social media hours show no statistically significant association with any mental health outcome when analyzed as a continuous predictor (Table 5).

Table 5: Pearson Correlations: Daily Social Media Hours vs. Mental Health Outcomes

Outcome	r	p -value
Stress level	+0.031	0.288
Anxiety level	+0.028	0.335
Addiction level	-0.025	0.388
Sleep hours	-0.010	0.743
Academic performance	+0.013	0.648

One-way ANOVA across Low, Medium, and High usage bins similarly fails to reach conventional significance thresholds (Table 6).

Table 6: One-Way ANOVA: Mental Health Outcomes by Social Media Usage Bin

Outcome	<i>F</i>	<i>p</i> -value	Sig.
Stress level	2.80	0.061	(marginal)
Anxiety level	1.02	0.362	n.s.
Addiction level	2.41	0.091	n.s.

5.2 Usage Bin Means

Table 7 shows that mean stress and anxiety levels are remarkably stable across usage tiers, further undermining the dose-response hypothesis.

Table 7: Mean Outcomes by Social Media Usage Bin

Bin	Stress	Anxiety	Addiction	Sleep (h)
Low (1–3 h)	5.51	5.45	5.74	6.54
Medium (3–5 h)	5.14	5.71	5.29	6.35
High (5–8 h)	5.61	5.71	5.63	6.45

5.3 Platform Moderation

While aggregate hours show no dose-response, platform type does produce modest differences. TikTok users report the highest mean anxiety (5.75) and show the steepest anxiety gradient at High usage (5.95). Multi-platform users (“Both”) report the highest mean stress (5.55). Instagram users fall in between on both measures.

Figures 3 and 4 visualize these patterns.

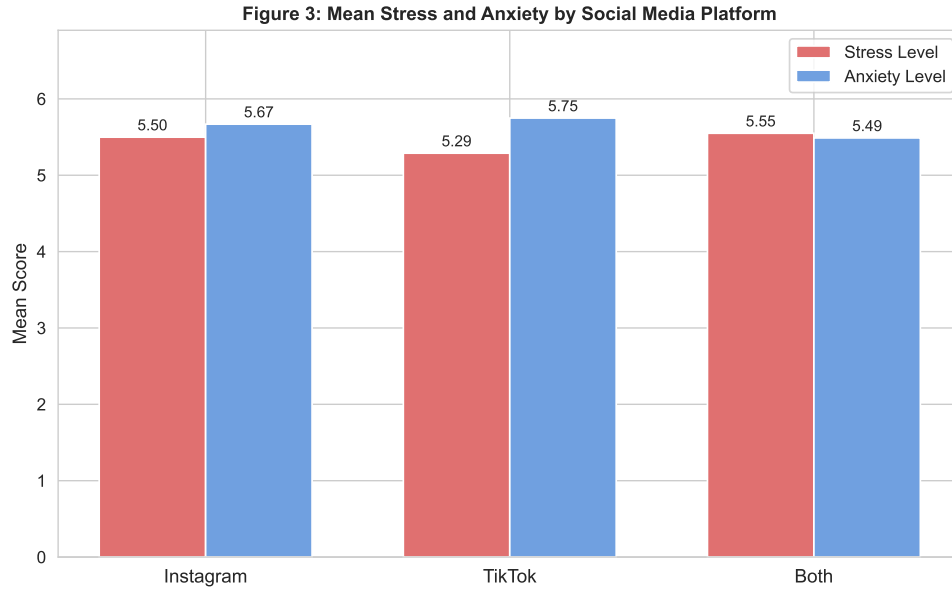


Figure 3: Mean stress and anxiety levels by social media platform. TikTok users show the highest mean anxiety; multi-platform (“Both”) users show the highest mean stress. Differences are modest but directionally consistent.

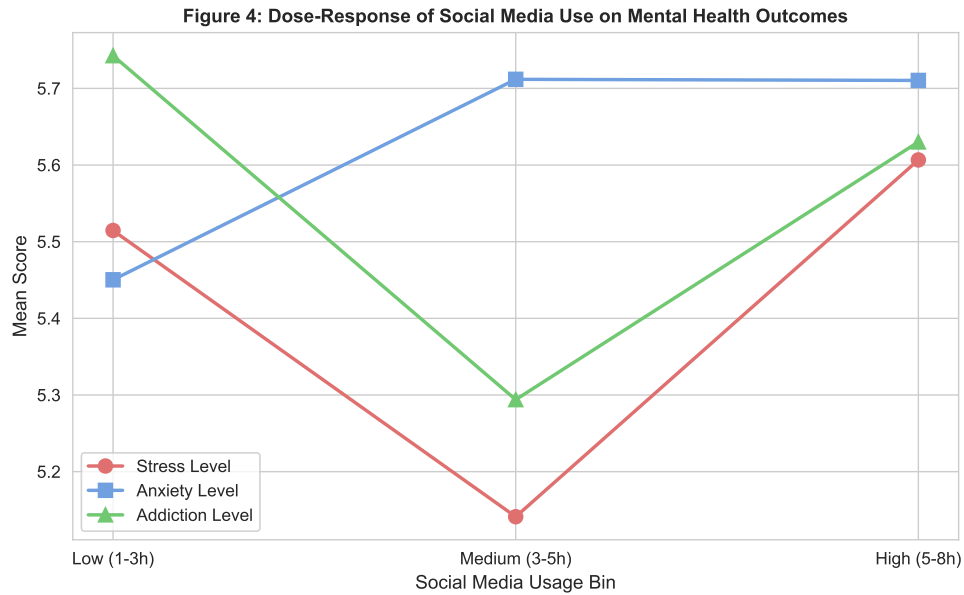


Figure 4: Dose-response of social media usage on mental health outcomes across Low (1–3 h), Medium (3–5 h), and High (5–8 h) daily usage bins. The near-flat trajectories for all three outcomes indicate no meaningful dose-response relationship in this sample.

Figure 5: Social Media Hours vs Mental Health Outcomes

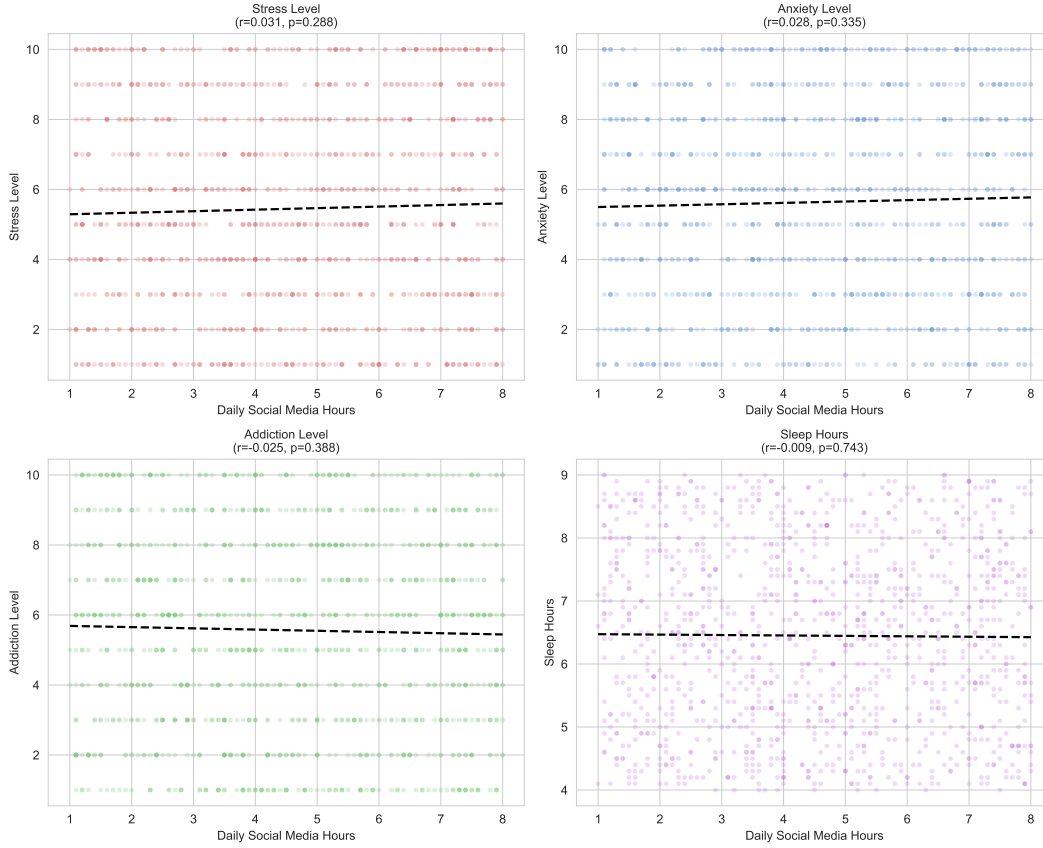


Figure 5: Scatter plots of daily social media hours against stress, anxiety, addiction, and sleep hours with OLS trend lines. The negligible slopes and near-zero r values confirm the absence of a linear dose-response association across all four outcomes.

6 RQ3: Sleep as a Protective Factor

6.1 Baron & Kenny Mediation Results

Table 8 presents the mediation coefficients for both stress and anxiety.

Table 8: Baron & Kenny Mediation Analysis: Screen Time $\rightarrow$ Sleep $\rightarrow$ Mental Health

Outcome	Path	Coefficient	SE	Interpretation
Stress	c ($X \rightarrow Y$, total)	-0.023	0.118	n.s.
	a ($X \rightarrow M$)	+0.019	0.058	n.s.
	b ($M \rightarrow Y \mid X$)	-0.022	0.058	n.s.
	c' ($X \rightarrow Y \mid M$, direct)	-0.023	0.118	n.s.
	Indirect ($a \times b$)	-0.0004	0.002	Sobel $z = -0.25$, n.s.
Anxiety	c ($X \rightarrow Y$, total)	-0.054	0.116	n.s.
	a ($X \rightarrow M$)	+0.019	0.058	n.s.
	b ($M \rightarrow Y \mid X$)	-0.024	0.057	n.s.
	c' ($X \rightarrow Y \mid M$, direct)	-0.053	0.116	n.s.
	Indirect ($a \times b$)	-0.0005	0.002	Sobel $z = -0.26$, n.s.

Conclusion: The mediation hypothesis is not supported. Path a (screen time $\rightarrow$ sleep) is non-significant ($r = 0.010$, $p = 0.72$), which violates the first prerequisite of mediation. Pre-sleep screen time does not measurably reduce sleep hours in this sample, so the mediation chain cannot be established.

6.2 Sleep Duration and Depression

Although the mediation model does not hold, sleep duration remains the most powerful predictor of depression (Section 4). The sleep-bin analysis reveals a stark non-linear pattern.

Table 9: Mental Health Outcomes by Sleep Duration Category

Sleep Category	n	Stress (mean)	Anxiety (mean)	Depression %
Short (<6 h)	480	5.50	5.69	6.5%
Adequate (6–8 h)	514	5.46	5.55	0.0%
Long (>8 h)	206	5.30	5.75	0.0%

All 31 depression cases occur in the short-sleep group (<6 h), yielding a depression prevalence of 6.5% in this subgroup compared with 0% in teens sleeping ≥ 6 h. This represents a clean threshold effect rather than a graded dose-response.

Importantly, short-sleep teens do not differ meaningfully in stress or anxiety from adequate-sleep teens (5.50 vs. 5.46 and 5.69 vs. 5.55, respectively), suggesting that sleep’s role in depression is not channeled through conscious stress or anxiety experiences but through other pathways (e.g., neurobiological, affective dysregulation).

Figures 6 and 7 display the mediation scatter panels and sleep-bin bar chart.

Figure 6: Mediation Pathway — Screen Time → Sleep → Mental Health

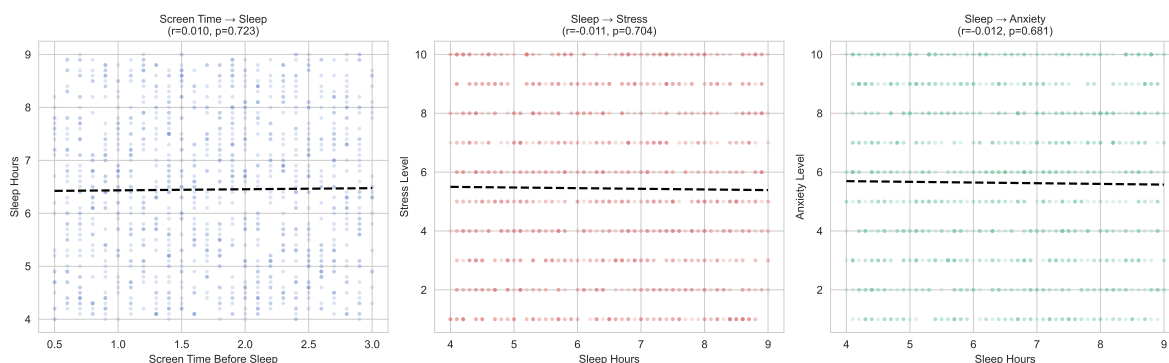


Figure 6: Mediation pathway scatter plots: (A) pre-sleep screen time vs. sleep hours ($r = 0.010$, n.s.); (B) sleep hours vs. stress level ($r = -0.011$, n.s.); (C) sleep hours vs. anxiety level ($r = -0.012$, n.s.). The flat regression lines in panels (A)–(C) confirm that neither path a nor path b reaches significance, ruling out formal mediation.

Figure 7: Mental Health Outcomes by Sleep Duration Category

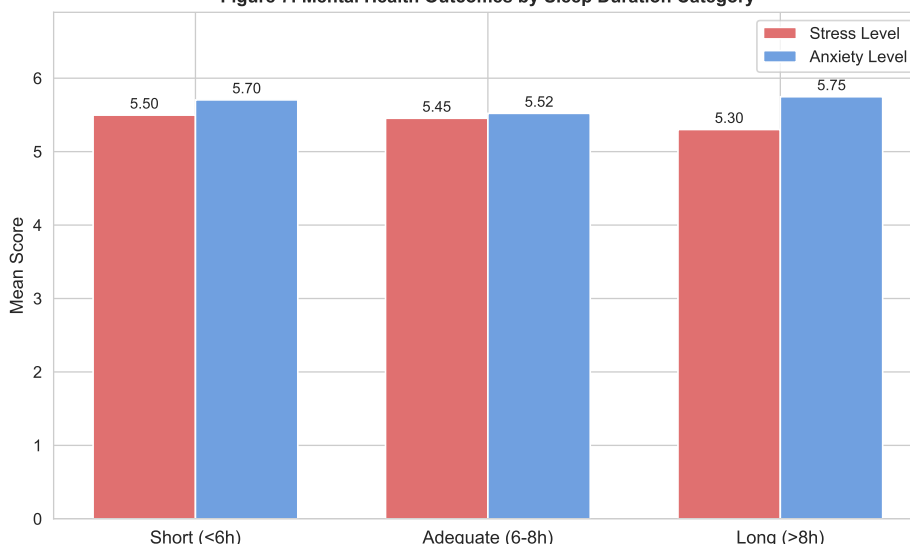


Figure 7: Mean stress and anxiety by sleep duration category. Levels are remarkably stable across Short, Adequate, and Long sleep groups, consistent with the non-significant mediation paths. Depression (not shown here) is exclusively concentrated in the Short (<6 h) group (6.5% vs. 0%), pointing to a threshold rather than graded relationship.

7 Discussion

7.1 Synthesis of Findings

Taken together, the three research questions paint a coherent picture. Sleep deprivation is the single most consequential behavioral factor for adolescent depression, operating as a threshold rather than as a mediator of other exposures. The anxiety-stress-social media triad forms a secondary risk cluster. That social media *hours* outperform addiction-level scores in the regression model

($\beta = 1.645$ vs. $\beta = 0.171$) may reflect measurement issues — the addiction scale may not adequately capture problematic usage — or may genuinely indicate that passive, high-volume exposure is more detrimental than compulsive short-burst use.

The null dose-response result for RQ2 is notable. The absence of a linear hourly effect on stress and anxiety challenges the intuition that “more screen time = worse mental health,” at least in cross-sectional data. Platform type provides a more nuanced story: TikTok’s algorithmically curated short-form video feed appears to be associated with higher anxiety (particularly at high usage), while multi-platform simultaneous use correlates with higher stress.

The failure to detect mediation via sleep in RQ3 does not diminish sleep’s importance — quite the opposite. The threshold finding (zero depression at ≥ 6 h, 6.5% at < 6 h) suggests that adequate sleep may act as a *moderator* or a *prerequisite* for resilience rather than as a simple pathway variable. Future studies should test this moderation model explicitly.

7.2 Limitations

Several limitations warrant caution.

1. **Cross-sectional design.** All three analyses use cross-sectional data, precluding causal inference. The observed associations between social media hours and depression may reflect reverse causality (depressed teens using media to cope).
2. **Class imbalance.** With only 31 depression cases (2.6%), the logistic model may be overfitting to the majority class despite high AUC. The impressive accuracy is partly inflated by the base rate. A balanced training set or cost-sensitive learning would strengthen RQ1 conclusions.
3. **Self-report bias.** Social media hours, sleep duration, and stress/anxiety levels are all self-reported, likely introducing recall bias and social desirability effects.
4. **Platform heterogeneity.** “Both” (multi-platform) is a heterogeneous category that conflates different usage intensities. Disaggregating by specific platform combinations would sharpen the platform moderation analysis.
5. **Omitted variables.** Family environment, socioeconomic status, prior mental health history, and school-related stressors are not included in the dataset and represent important confounders.

8 Conclusion

This study asked three empirical questions about teen mental health. The answers are:

1. **RQ1 — Best predictors of depression:** Sleep hours (protective; OR = 0.111), anxiety (OR = 6.50), stress (OR = 5.68), and daily social media hours (OR = 5.18) jointly explain adolescent depression risk with near-perfect discrimination (AUC = 0.9936, McFadden R^2 = 0.688). Sleep deprivation is the single largest modifiable risk factor.
2. **RQ2 — Social media dose-response:** No significant linear dose-response exists between daily social media hours and stress, anxiety, or addiction ($|r| \leq 0.031$, all $p > 0.06$). However, platform choice is a meaningful moderator: TikTok users show the highest anxiety and multi-platform users show the highest stress.

3. **RQ3 — Sleep as mediator:** Formal Baron & Kenny mediation of screen time effects through sleep is not supported (path a : $r = 0.010$, $p = 0.72$). Nonetheless, sleeping fewer than 6 hours per night is associated with a 6.5% depression prevalence compared with 0% for adequate sleepers — a finding with clear public health implications.

From a policy standpoint, interventions that protect adolescent sleep (e.g., delayed school start times, device curfews, parental guidance on nighttime phone use) hold the greatest promise for reducing depression risk. Social media restriction policies may be better targeted at specific high-anxiety platforms (TikTok) and multi-platform simultaneous use rather than broad hourly limits.

Future work should employ longitudinal designs, objective sleep actigraphy, and richer platform-specific engagement metrics to disentangle causality from correlation and to validate these cross-sectional findings at scale.

Reproducibility

All analyses in this paper are fully reproducible. The Jupyter notebook `research_notebook.ipynb` (located in the same output directory as this paper) re-generates all figures and tables from the raw dataset `Teen_Mental_Health_Dataset.csv`. Run:

```
jupyter nbconvert --to notebook --execute research_notebook.ipynb \
    --output research_notebook_executed.ipynb
```

Software versions: Python 3.9+, pandas, scikit-learn, scipy, matplotlib, seaborn.